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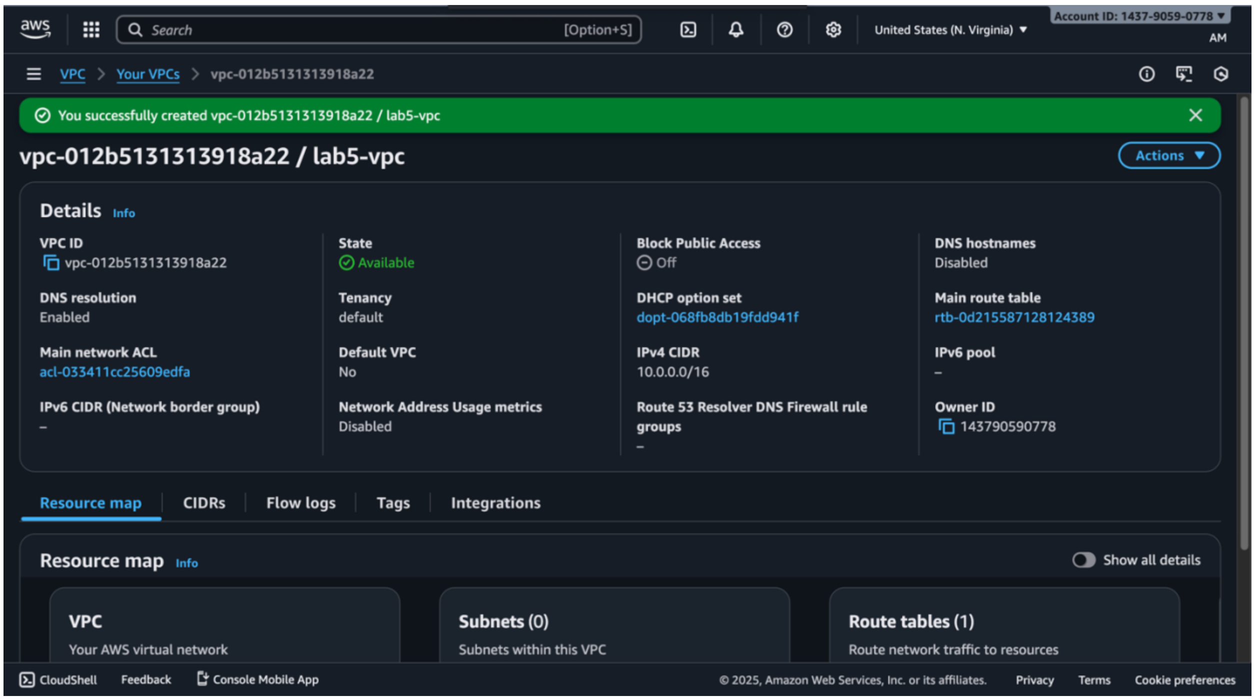
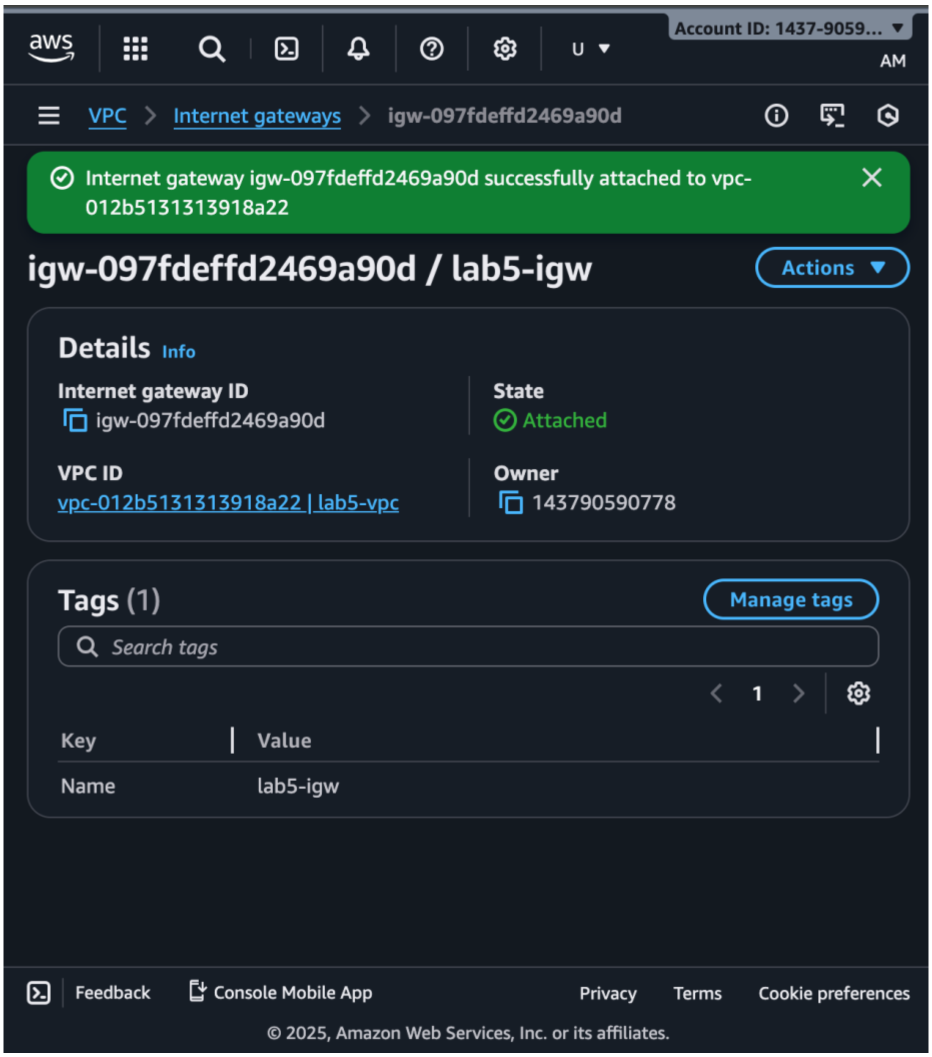
Lab5

11/05/2025

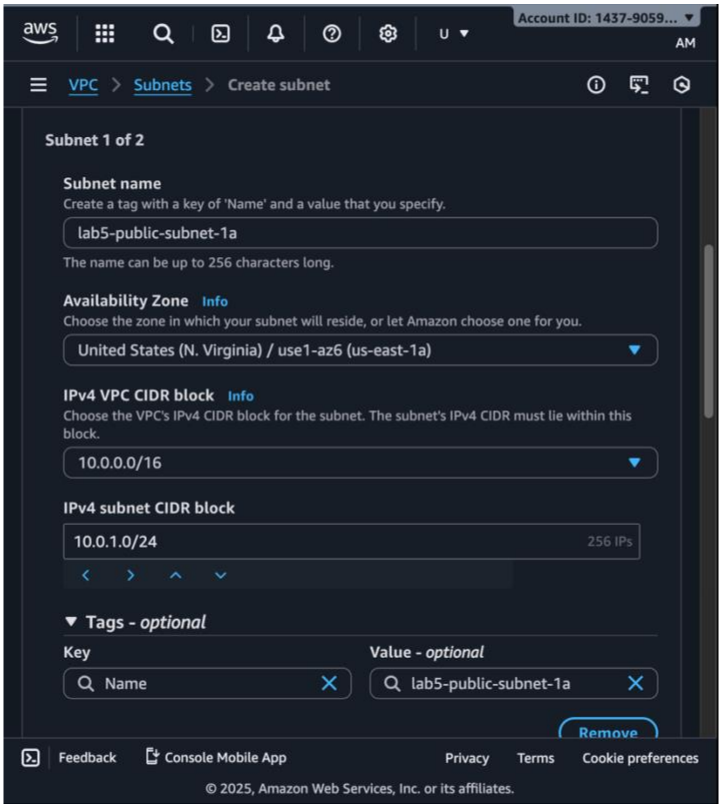
Creating a VPC: Define an isolated network by creating a Virtual Private Cloud (VPC). This provides complete control over routing, subnets, and IP addresses. Every resource in this lab has private IP space reserved by the CIDR block 10.0.0.0/16.

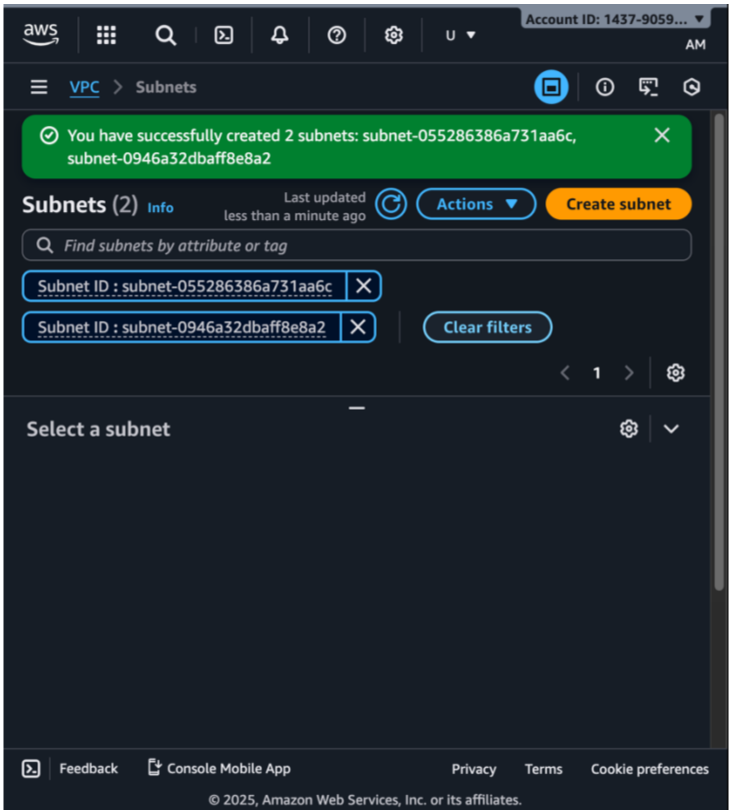


Creating an internet gateway: The VPC is connected to the public internet via an Internet Gateway (IGW). Once routing and security rules are set up, attaching it to the VPC enables resources like EC2 instances to transmit and receive traffic outside of AWS.



Creating Subnets: In separate Availability Zones (us-east-1a and us-east-1b), two public subnets are established. This increases fault tolerance and makes it possible for the load balancer to divide traffic among zones for high availability.




Account ID: 1437-9059...
AM
VPC > Subnets
You have successfully created 2 subnets: subnet-055286386a731aa6c, subnet-0946a32dbaff8e8a2
Subnets (2) Info
Last updated less than a minute ago
Actions
Create subnet
Find subnets by attribute or tag
Subnet ID : subnet-055286386a731aa6c
Subnet ID : subnet-0946a32dbaff8e8a2
Clear filters
1
Select a subnet
Feedback
Console Mobile App
Privacy
Terms
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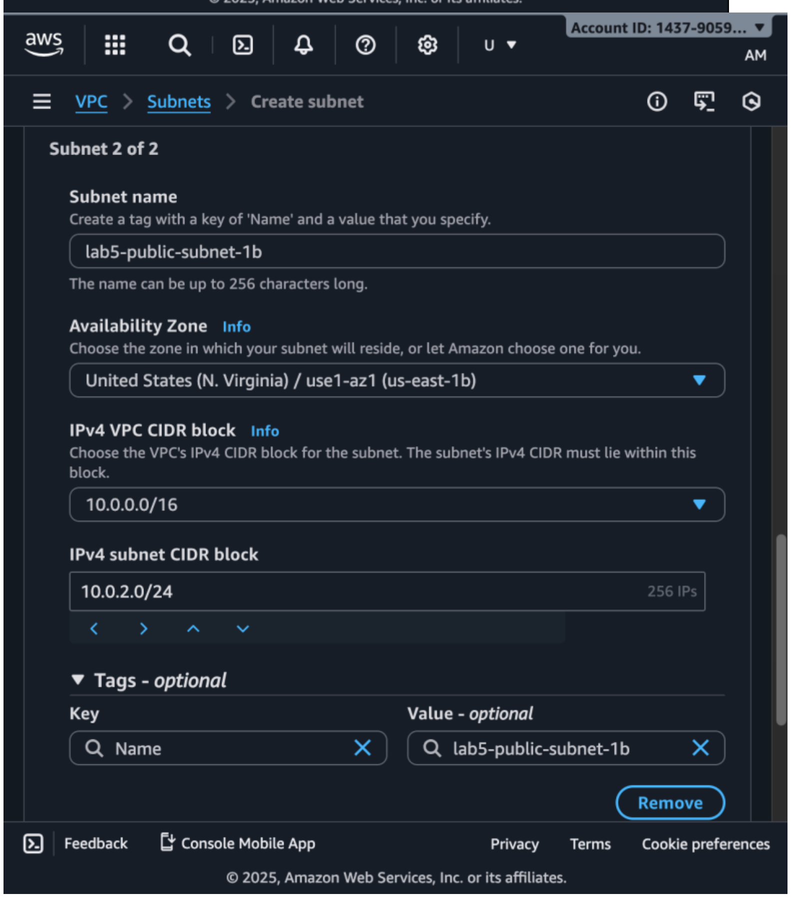
Account ID: 1437-9059...
AM
VPC > Subnets > Create subnet
Subnet 2 of 2
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
lab5-public-subnet-1b
The name can be up to 256 characters long.
Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az1 (us-east-1b)
IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16
IPv4 subnet CIDR block
10.0.2.0/24
256 IPs
Tags - optional
Key
Name
Value - optional
lab5-public-subnet-1b
Remove
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
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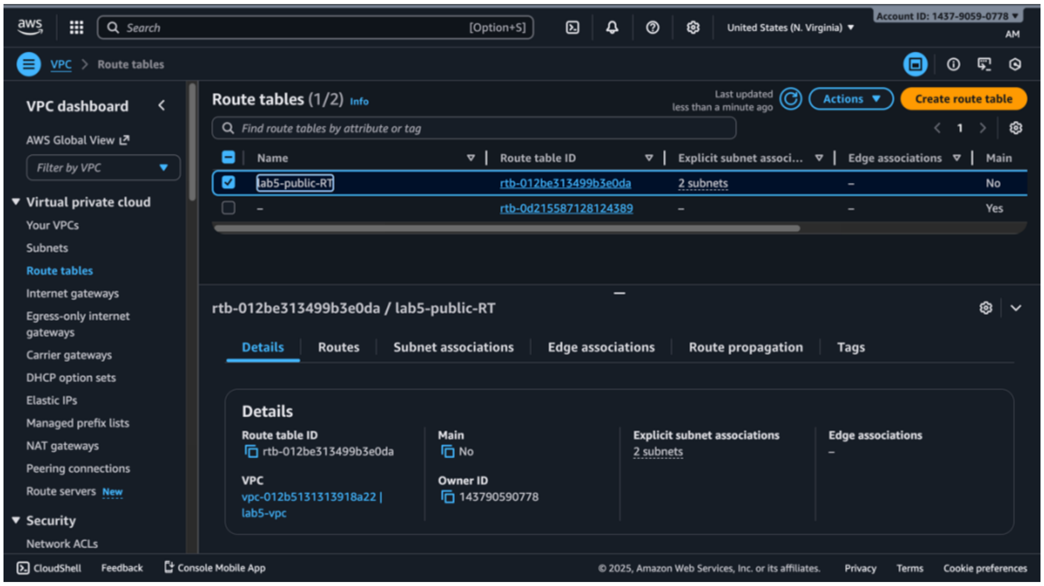
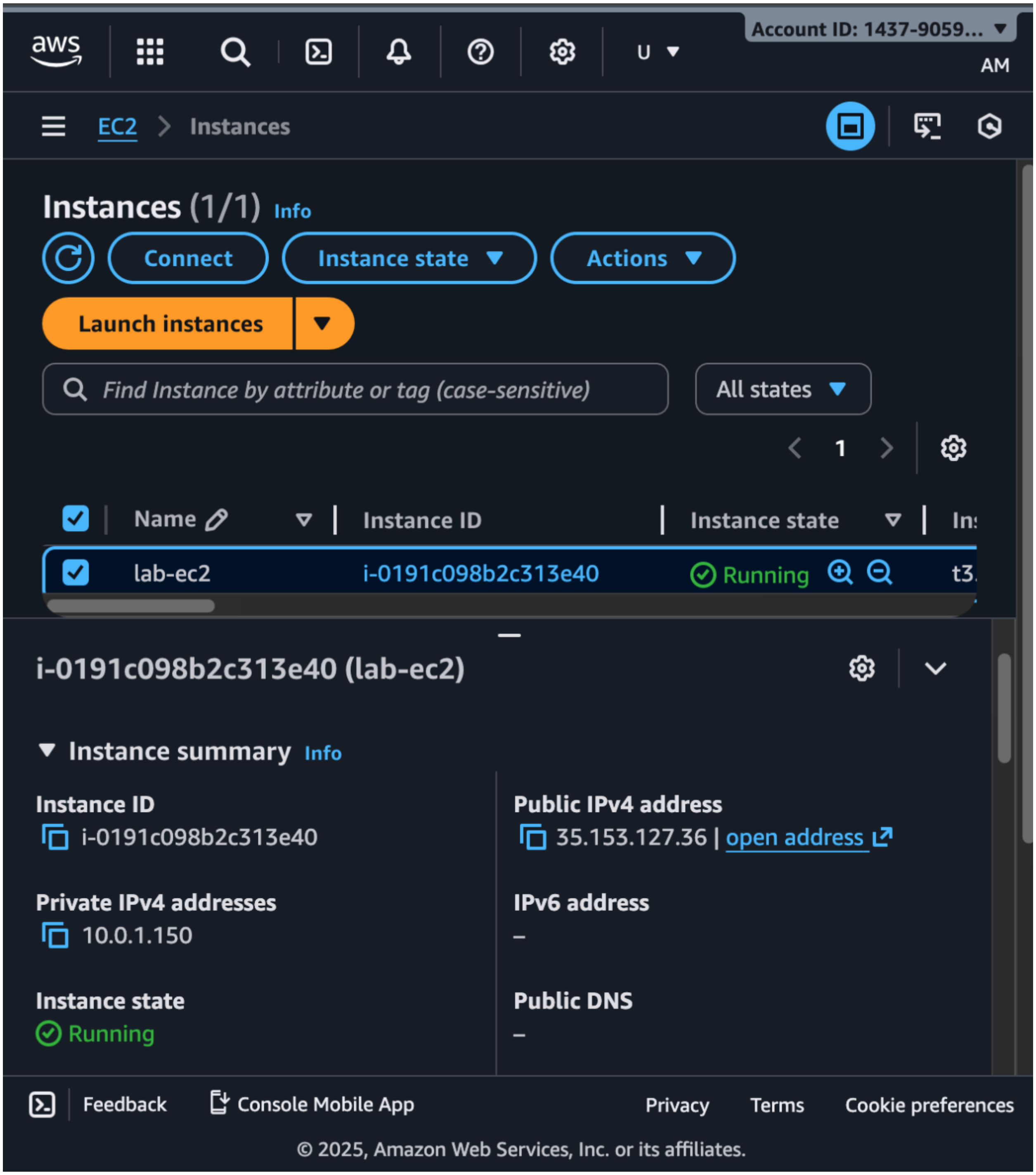
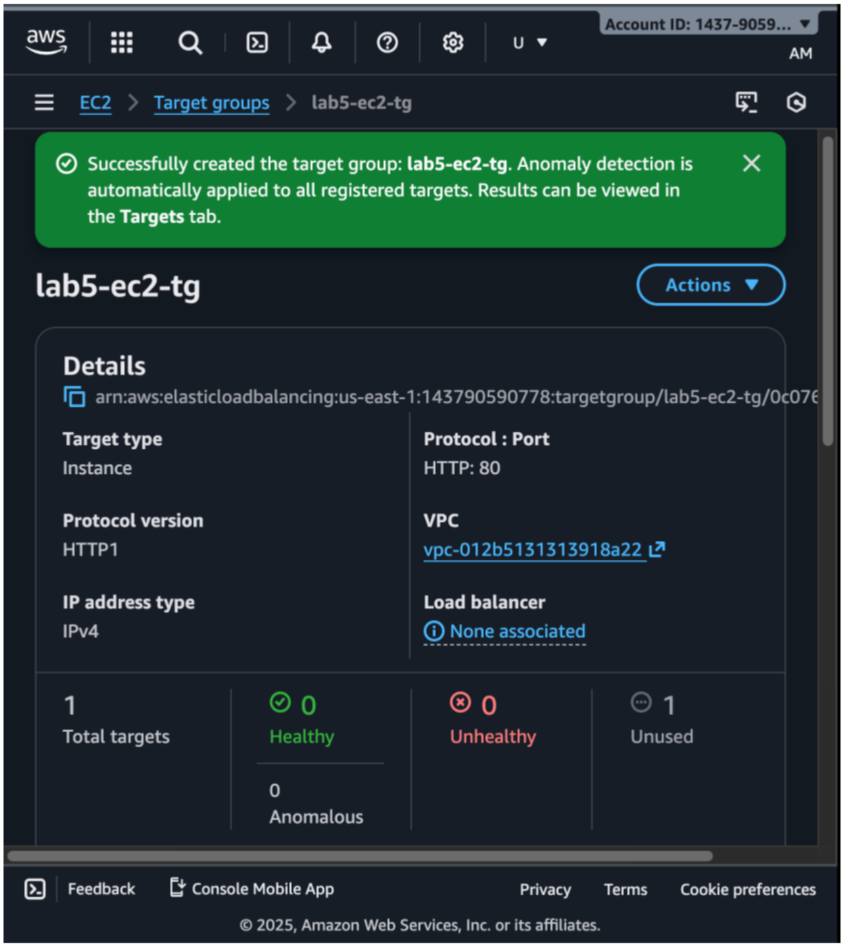
Creating a Route table: Network traffic is directed by the route table. Instances in these subnets will be able to access the internet if both subnets are associated and a route 0.0.0.0/0 pointing to the IGW is added.



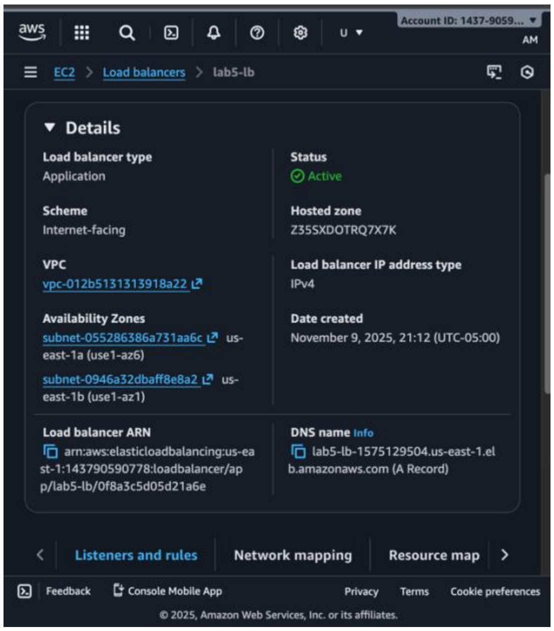
Creating an EC2 Instance: A basic web server is an Ubuntu EC2 instance. Apache 2 installs automatically using user data and displays a simple HTML page with the hostname and IP. Afterwards, the load balancer uses this instance as a backend target.



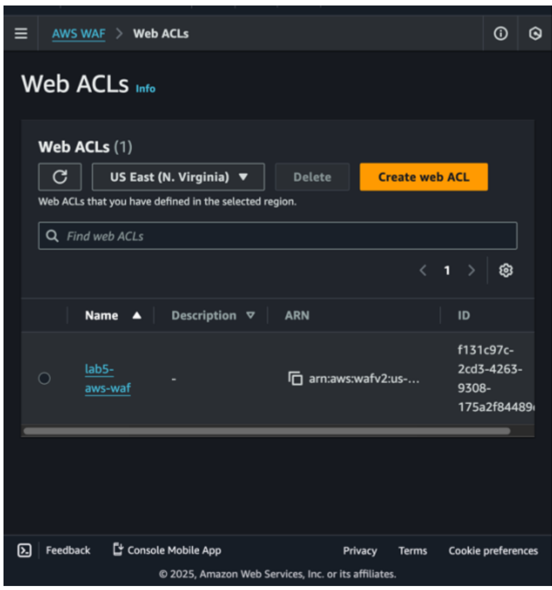
Creating a Target Group: EC2 instances are logically grouped for the load balancer by a target group. It routes HTTP traffic to them and keeps an eye on their health. The target in this case is the single instance lab5-ec2.

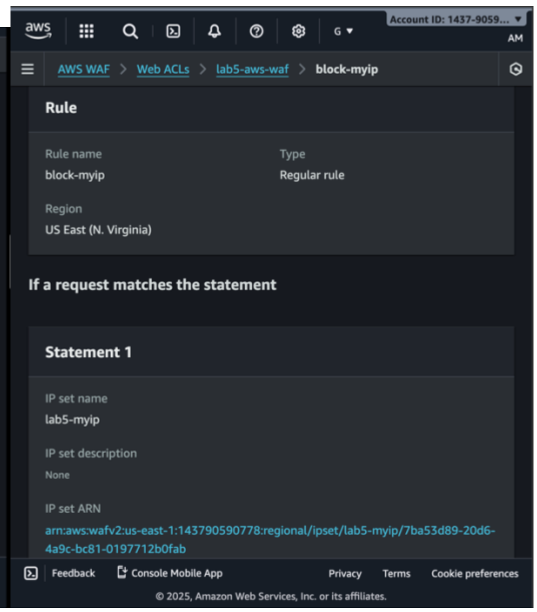


Creating a Load Balancer: The Application Load Balancer (ALB) distributes incoming HTTP requests across targets in multiple subnets. It increases availability and provides a single DNS endpoint for clients to reach the backend servers securely.

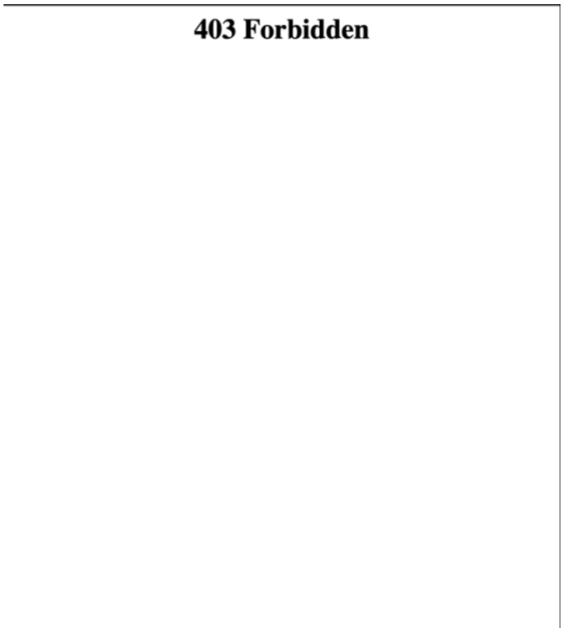


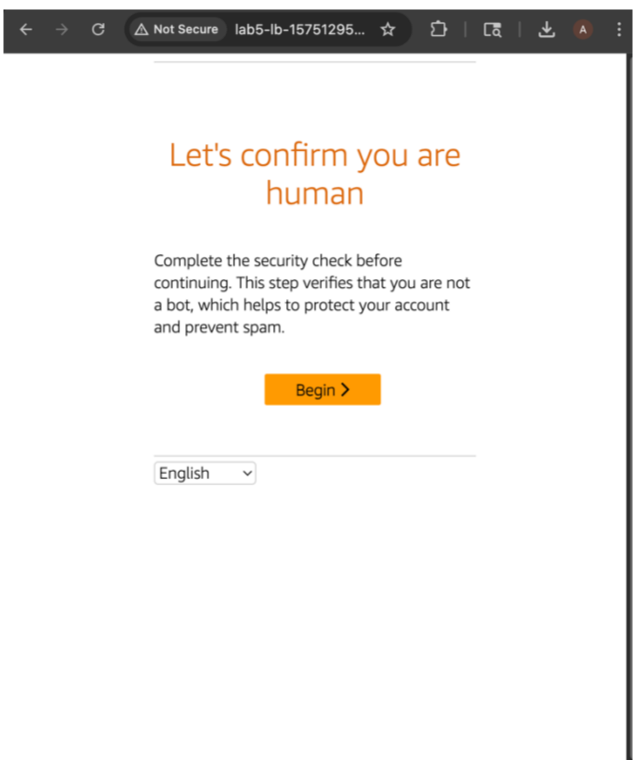
Creating a WAF: AWS WAF filters and manages HTTP requests to provide an additional degree of security. In this lab, a Web ACL with a rule preventing traffic from a particular IP set is made and connected to the ALB. This illustrates how WAF can employ rules to ban, permit, or challenge (CAPTCHA) users.



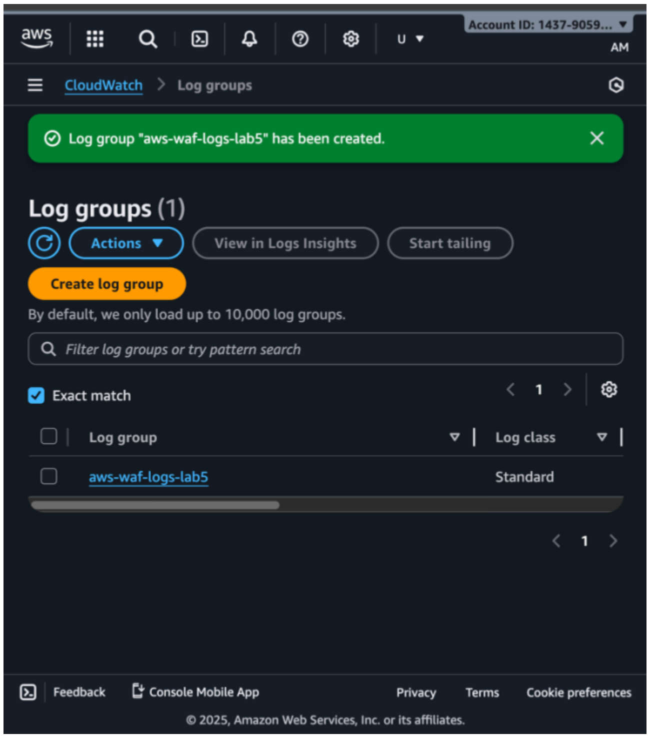


Verifying the Working of WAF: The WAF configuration is tested by gaining access to the ALB's DNS name. WAF is actively blocking the specified IP, as seen by the "403 Forbidden" response. Modifying the rule action to Allow and CAPTCHA demonstrates dynamic control over the processing of requests.





Creating CloudWatch Logs: CloudWatch Logs collects and stores WAF activity for analysis. Creating a log group like aws-waf-logs-lab5 lets administrators monitor requests, troubleshoot security events, and visualize blocked or allowed traffic patterns.



Attaching and Viewing CloudWatch Logs to the WAF: Enabling logging on the WAF sends all web ACL events to CloudWatch. Viewing the log streams verifies that WAF actions are recorded, proving end-to-end monitoring and auditing are active.